

七. 隐函数的微分法

1. 定理1: 设函数 $F(x, y)$ 在点 $P(x_0, y_0)$ 的某一邻域内满足 ① 具有连续的偏导数

② $F(x_0, y_0) = 0$ ③ $F_y(x_0, y_0) \neq 0$

平面穿过 $z=0$

则方程 $F(x, y) = 0$ 在点 x_0 的某邻域内可唯一确定一个单值连续函数 $y = f(x)$, 满足条件

$y_0 = f(x_0)$ 并有连续导数 $\frac{dy}{dx} = -\frac{F_x}{F_y}$

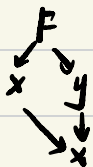


设 $y = f(x)$ 为方程 $F(x, y) = 0$ 所确定的隐函数, 则 $\frac{dy}{dx} =$

$F(x, f(x)) = 0$

$F_x + F_y \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{F_x}{F_y}$



$dF(x, y) = 0$

$F_x dx + F_y dy = 0$

$\frac{dy}{dx} = -\frac{F_x}{F_y}$

2. 定理1: 设函数 $F(x, y, z)$ 在点 $P(x_0, y_0, z_0)$ 的某一邻域内满足 ① 具有连续的偏导数

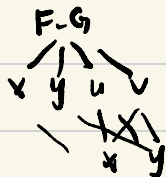
② $F(x_0, y_0, z_0) = 0$ ③ $F_z(x_0, y_0, z_0) \neq 0$

则方程 $F(x, y, z) = 0$ 在点 x_0 的某邻域内可唯一确定一个单值连续函数 $z = f(x, y)$, 满足条件

$z_0 = f(x_0, y_0)$ 并有连续导数 $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$ $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$

3. 隐函数方程组

$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases} \rightarrow \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$



$$\begin{cases} F_x + F_u \frac{\partial u}{\partial x} + F_v \frac{\partial v}{\partial x} = 0 \\ G_x + G_u \frac{\partial u}{\partial x} + G_v \frac{\partial v}{\partial x} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial x} \end{cases}$$

$$4. \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases} \Rightarrow \begin{cases} y = y(x) \\ z = z(x) \end{cases} \quad \text{可求 } \frac{dy}{dx}, \frac{dz}{dx}$$

八. 方向导数与梯度

1. 方向导数: 若函数 $f(x, y, z)$ 在点 $P(x, y, z)$ 处沿方向 l 方向角为 α, β, γ 存在下列极限: $\lim_{\rho \rightarrow 0^+} \frac{\Delta f}{\rho} = \lim_{\rho \rightarrow 0^+} \frac{f(x+\Delta x, y+\Delta y, z+\Delta z) - f(x, y, z)}{\rho} \stackrel{\text{记作}}{=} \frac{\partial f}{\partial l} \Big|_{P(x, y, z)}$

$$\begin{cases} \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = |\rho \mathbf{p}| \\ \Delta x = \rho \cos \alpha, \Delta y = \rho \cos \beta, \Delta z = \rho \cos \gamma \end{cases}$$

则称 $\frac{\partial f}{\partial l}$ 为函数在点 P 沿 l 方向的方向导数

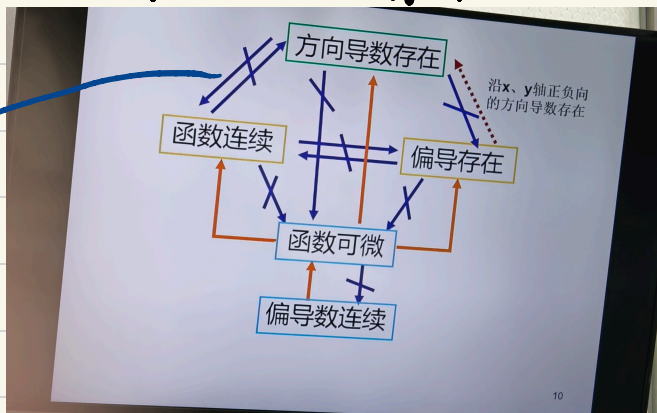
前提条件

2. 定理: 若函数 $f(x, y, z)$ 在点 $P(x, y, z)$ 处可微, 则函数在该点沿 l 方向的方向导数存在, 且有 $\frac{\partial f}{\partial l} = \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta + \frac{\partial f}{\partial z} \cos \gamma$, 其中 α, β, γ 为 l 方向角

3. 对于二元函数: $\frac{\partial f}{\partial l} \Big|_{P(x, y)} = \lim_{\rho \rightarrow 0^+} \frac{f(x+\Delta x, y+\Delta y) - f(x, y)}{\rho} = \lim_{\rho \rightarrow 0^+} \frac{f(x+\rho \cos \alpha, y+\rho \cos \beta) - f(x, y)}{\rho}$

f 在 P 点可微 $\Rightarrow \frac{\partial f}{\partial l} \Big|_{P(x, y)} = f_x \cos \alpha + f_y \cos \beta$

$$z = \begin{cases} 1 & y = x^2, x \neq 0 \\ 0 & \text{其它} \end{cases}$$



九、梯度

$$\text{方向导数公式 } \frac{\partial f}{\partial l} = \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta + \frac{\partial f}{\partial z} \cos \gamma$$

$$\downarrow \quad \text{令向量 } \vec{G} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\vec{e} = (\cos \alpha, \cos \beta, \cos \gamma)$$

$$\frac{\partial f}{\partial l} = \vec{G} \cdot \vec{e} = |\vec{G}| \cdot \cos(\vec{G}, \vec{e}) \quad (|\vec{e}| = 1)$$

1. 定义: 向量 $\vec{G}$ 称为函数 $f(x, y, z)$ 在点 P 处的梯度, 记作 $\text{grad} f$ 即

$$\nabla f = \text{grad} f = \left\{ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\} \quad \text{函数增长最快的向量}$$

2. 对函数 $z = f(x, y)$, 曲线 $\begin{cases} z = f(x, y) \\ z = c \end{cases}$ 在 xy 面上的投影 $L^* = \{x, y | z = c\}$ 称为函数

f 的等值线

设 f_x, f_y 不同时为 0, 则 L^* 上点 P 处的法向量为 $\|f_x, f_y\|_P = \text{grad} f|_P$

十、空间曲线的切线与法平面

1. 曲线方程为参数方程的情况

$$\Gamma: x = \varphi(t), y = \psi(t), z = w(t)$$

设 $t = t_0$ 对应 $M(x_0, y_0, z_0)$

$$t = t_0 + \Delta t \text{ 对应 } M'(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$$

$$MM': \frac{x-x_0}{\frac{\Delta x}{\Delta t}} = \frac{y-y_0}{\frac{\Delta y}{\Delta t}} = \frac{z-z_0}{\frac{\Delta z}{\Delta t}}$$

$M' \rightarrow M \quad \Delta t \rightarrow 0$ 时

$$\text{切线方程: } \frac{x-x_0}{\varphi'(t_0)} = \frac{y-y_0}{\psi'(t_0)} = \frac{z-z_0}{w'(t_0)}$$

曲线的切向量 $T = \{\varphi'(t_0), \psi'(t_0), w'(t_0)\}$

法平面的方程: $\varphi'(t_0)(x-x_0) + \psi'(t_0)(y-y_0) + w'(t_0)(z-z_0) = 0$

2. 曲线为一般式情况

$$\text{光滑曲线 } \Gamma: \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases} \Rightarrow \begin{cases} x = x \\ y = y(x) \\ z = z(x) \end{cases}$$

$$\text{切线: } \frac{x-x_0}{1} = \frac{y-y_0}{y'(x_0)} = \frac{z-z_0}{z'(x_0)}$$

十一: 曲面的切平面和法线

设有光滑曲面 $\Sigma: F(x, y, z) = 0$

通过其上定点 $M(x_0, y_0, z_0)$ 任意引一条光滑曲线 $\Gamma: x=\varphi(t), y=\psi(t), z=w(t)$, 设

$t=t_0$ 对应点 M , 且 $\varphi'(t_0), \psi'(t_0), w'(t_0)$ 不全为 0, 则 Γ 在点 M 的切向量为 $\vec{\tau} = (\varphi'(t_0), \psi'(t_0), w'(t_0))$

$$\text{切线方程: } \frac{x-x_0}{\varphi'(t_0)} = \frac{y-y_0}{\psi'(t_0)} = \frac{z-z_0}{w'(t_0)}$$

曲面 Σ 在点 M 的法向量 $\vec{n} = (F_x(x_0, y_0, z_0), F_y(x_0, y_0, z_0), F_z(x_0, y_0, z_0))$

切平面 $F_x(x-x_0) + F_y(y-y_0) + F_z(z-z_0) = 0$

$$\text{法线: } \frac{x-x_0}{F_x} = \frac{y-y_0}{F_y} = \frac{z-z_0}{F_z}$$

特别: $z = f(x, y)$

$$f(x, y, z) = f(x, y), \quad z = 0$$

$$\text{切平面 } f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$

十二. 多元函数的泰勒公式

记号: $\cdot (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}) f(x_0, y_0)$ 表示 $h f_x(x_0, y_0) + k f_y(x_0, y_0)$

$$\cdot (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^2 f(x_0, y_0) \text{ 表示 } h^2 \frac{\partial^2 f(x_0, y_0)}{\partial x^2} + 2hk \frac{\partial^2 f(x_0, y_0)}{\partial x \partial y} + k^2 \frac{\partial^2 f(x_0, y_0)}{\partial y^2}$$

定理1: 设 $z = f(x, y)$ 在点 (x_0, y_0) 的某邻域有直到 $n+1$ 阶连续偏导数, (x_0+h, y_0+k) 为此邻域内任一点, 则有

$$\begin{aligned} f(x_0+h, y_0+k) &= f(x_0, y_0) + (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}) f(x_0, y_0) + \frac{1}{2!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^2 f(x_0, y_0) \\ &\quad + \dots + \frac{1}{n!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(x_0, y_0) + \frac{1}{(n+1)!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^{n+1} f(x_0+h, y_0+\theta k) \end{aligned}$$

十三. 多元函数的极值

定义: 若函数 $z = f(x, y)$ 在点 (x_0, y_0) 的某邻域内有 $f(x, y) \leq f(x_0, y_0)$ 或 $f(x, y) \geq f(x_0, y_0)$, 则称函数在该点取极大值(极小值)

1. 定理1: (必要条件) 函数 $z = f(x, y)$ 在点 (x_0, y_0) 存在偏导数, 且在该点取得极值, 则有 驻点 $f'_x(x_0, y_0) = 0, f'_y(x_0, y_0) = 0$

驻点不一定是极值点, eg: $z = xy$

2. 黑塞(Hesse)矩阵: $Hf(P_0) = \begin{bmatrix} f_{xx}(P_0) & f_{xy}(P_0) \\ f_{yx}(P_0) & f_{yy}(P_0) \end{bmatrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} P_0$

定理2: (极值的充分条件)

设 $P_0(x_0, y_0)$ 是 f 在 D 内的驻点, f 在 D 内有二阶连续偏导数, 记 $A = f_{xx}(P_0)$,

$$B = f_{xy}(P_0), C = f_{yy}(P_0), \Delta = AC - B^2$$

ii) $\Delta > 0, A < 0 \Rightarrow P_0$ 是 f 的极大值.

iii) $\Delta < 0 \Rightarrow P_0$ 不是 f 的极值点.

iv) $\Delta = 0 \Rightarrow$ 不能确定.

十五、条件极值

对自变量除定义域限制外,还有其它条件限制

1. 模型1: 求函数 $z=f(x, y)$ 在条件 $\varphi(x, y)=0$ 下的极值

① 方法1: 代入法

从 $\varphi(x, y) = 0$ 解出 $y = \psi(x) \rightarrow$ 求一元函数 $z = f(x, \psi(x))$ 的自由极值

② 方法2: 拉格朗日乘数法.

设 $z=f(x,y)$ 在 (x_0, y_0) 取得极值, 则 $z=f(x, y|u=1)$ 在 $x=x_0$ 取得极值.

$$\left. \begin{aligned} \frac{dz}{dx} = f_x + f_y \frac{dy}{dx} = 0 &\Rightarrow \frac{dy}{dx} = -\frac{f_x}{f_y} \\ \text{由 } \varphi(x, y) = 0 &\Rightarrow \frac{dy}{dx} = -\frac{\varphi_x}{\varphi_y} \end{aligned} \right\} \Rightarrow \frac{f_x}{f_y} = \frac{\varphi_x}{\varphi_y} \Rightarrow \nabla f(x_0, y_0) // \nabla \varphi(x_0, y_0) \Rightarrow \nabla f = -\lambda \nabla \varphi \Rightarrow \nabla(f + \lambda \varphi) = 0$$

$$F = f(x, y) + \lambda \varphi(x, y) \rightarrow \text{拉格朗日函数}$$

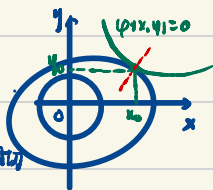
极值点满足

$$\begin{cases} F_x = f_x + \lambda \varphi_x = 0 \\ F_y = f_y + \lambda \varphi_y = 0 \\ F_\lambda = \varphi(x, y) = 0 \end{cases}$$

③ 方法3: 几何法

$$\begin{cases} \nabla f(x_0, y_0) // \nabla \varphi(x_0, y_0) \\ \varphi(x_0, y_0) = 0 \end{cases}$$

注: $\varphi(x, y) = 0$ 与 $z = f(x, y)$ 等值线相切



Eg: 求函数 $z = x^2 + 2y^2$ 满足条件 $x^2 + y^2 = 1$ 的最值

① 拉格朗日乘数法: $F(x, y, \lambda) = x^2 + 2y^2 + \lambda(x^2 + y^2 - 1)$

$$\begin{cases} F_x = 2x + 2\lambda x = 2(\lambda + 1)x = 0 \\ F_y = 4y + 2\lambda y = 2(2 + \lambda)y = 0 \\ F_\lambda = x^2 + y^2 - 1 = 0 \end{cases}$$

$x = 0, y = \pm 1$ 或 $x = \pm 1, y = 0 \rightarrow$ 驻点

② 几何法: $\{2x, 4y\} // \{2x, 2y\}$

$$\begin{cases} \frac{x}{2y} = \frac{x}{y} \Rightarrow xy = 0 \\ x^2 + y^2 = 1 \end{cases}$$

2. 模型2: 求函数 $u = f(x, y, z)$ 在 $\varphi(x, y, z) = 0$ 条件下极值

① $\varphi(x, y, z) = 0 \rightarrow z = z(x, y)$

设 $u = f(x, y, z(x, y))$ 在 (x_0, y_0) 取得极值.

$$u_x = f_x + f_z z_x = 0 = \{f_x, f_y, f_z\} \cdot \{1, 0, z_x\}$$

$$u_y = f_y + f_z z_y = 0 = \{f_x, f_y, f_z\} \cdot \{0, 1, z_y\}$$

$$\nabla f // \{0, 1, z_y\} \times \{1, 0, z_x\} // \{z_x, z_y, -1\} // \nabla \varphi \Rightarrow \nabla f + \lambda \nabla \varphi = 0$$

$$F = f(x, y, z) + \lambda \varphi(x, y, z)$$

$$F_x = F_y = F_z = F_\lambda = 0 \Rightarrow \text{驻点}$$

$$\textcircled{2} \text{ 几何法: } \begin{cases} \nabla f(x_0, y_0, z_0) \parallel \nabla \varphi(x_0, y_0, z_0) \\ \varphi(x_0, y_0, z_0) = 0 \end{cases}$$

3. 模型3: 函数 $u = f(x, y, z)$ 在条件 $\varphi_1(x, y, z) = 0, \varphi_2(x, y, z) = 0$ 下的极值

$$\textcircled{1} \begin{cases} \varphi_1(x, y, z) = 0 \\ \varphi_2(x, y, z) = 0 \end{cases} \Rightarrow \begin{cases} y = y(x) \\ z = z(x) \end{cases}$$

设 $u = f(x, y(x), z(x))$ 在 $x = x_0$ 取得极值

$$\frac{du}{dx} = f_x + f_y y' + f_z z' = 0 = \nabla f \cdot (1, y', z')$$

$$\frac{d\varphi_1}{dx} = \nabla \varphi_1 \cdot (1, y', z') = 0$$

$$\frac{d\varphi_2}{dx} = \nabla \varphi_2 \cdot (1, y', z') = 0$$

$$\nabla f, \nabla \varphi_1, \nabla \varphi_2 \text{ 共面} \Rightarrow \Delta f + \lambda_1 \Delta \varphi_1 + \lambda_2 \Delta \varphi_2 = 0 \Rightarrow \nabla (f + \lambda_1 \varphi_1 + \lambda_2 \varphi_2) = 0$$

$$F = f(x, y, z) + \lambda_1 \varphi_1(x, y, z) + \lambda_2 \varphi_2(x, y, z)$$

$$F_x = F_y = F_z = F_{\lambda_1} = F_{\lambda_2} = 0$$

$$\textcircled{2} \begin{cases} [\nabla F + \nabla \varphi_1 + \nabla \varphi_2] = 0 \\ \varphi_1(x, y, z) = 0 \\ \varphi_2(x, y, z) = 0 \end{cases}$$